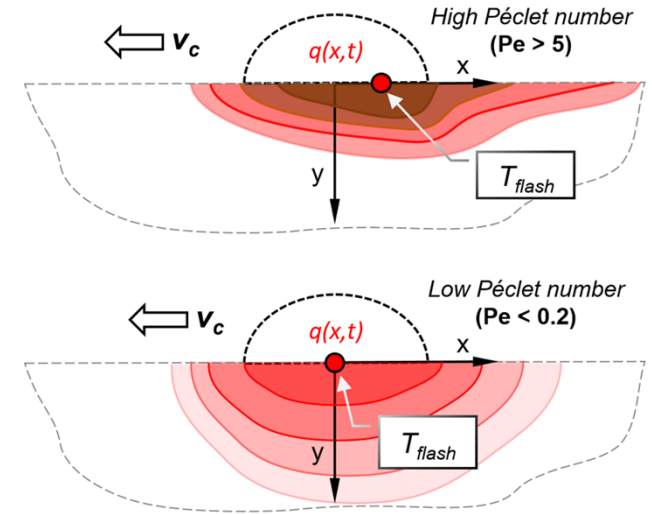




AER/ ME/MFS599/699

MODULE 2: MODELING OF THERMAL EFFECTS

Surface Integrity in Finishing Processes

Putting It All Together

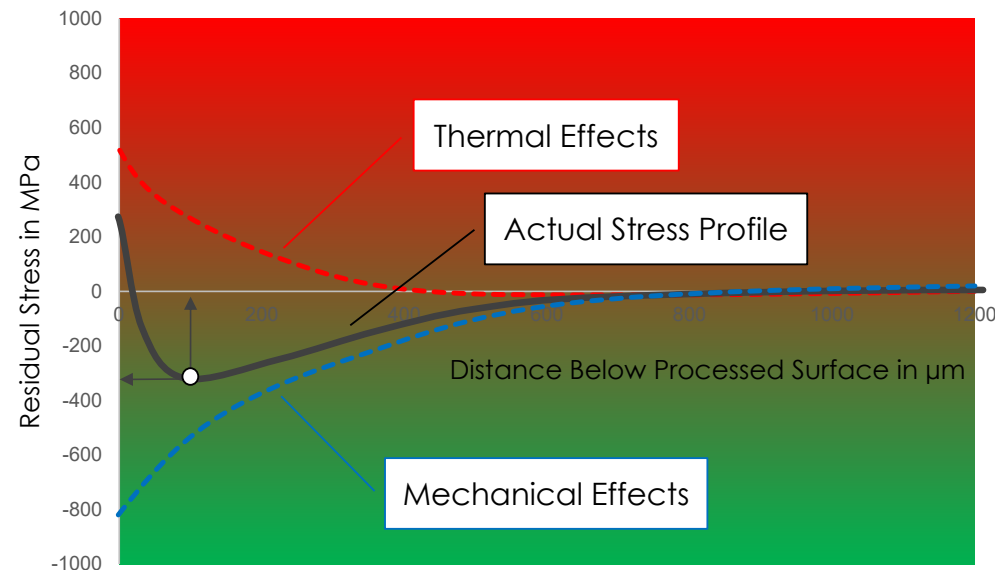
Surface Integrity is the result of Thermo-Mechanical Processing

- **Thermal Effects**

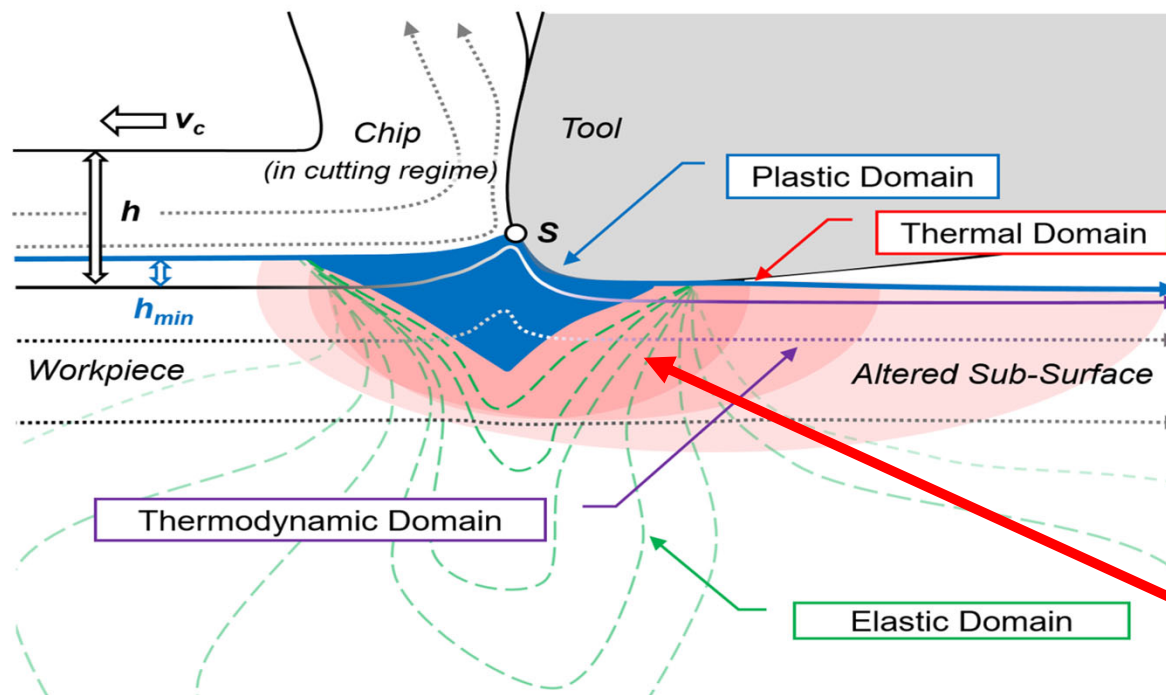
Can be engineered via cutting speed and advanced cooling/lubrication

- **Mechanical Effects**

Can be engineered by exploiting size effects (micro-geometries, etc.)

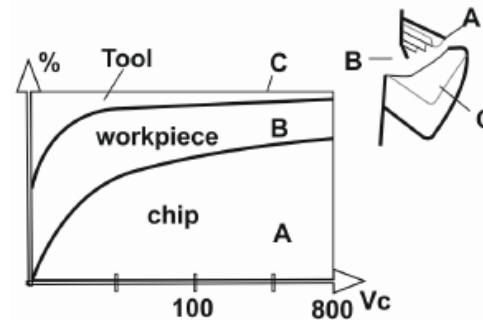
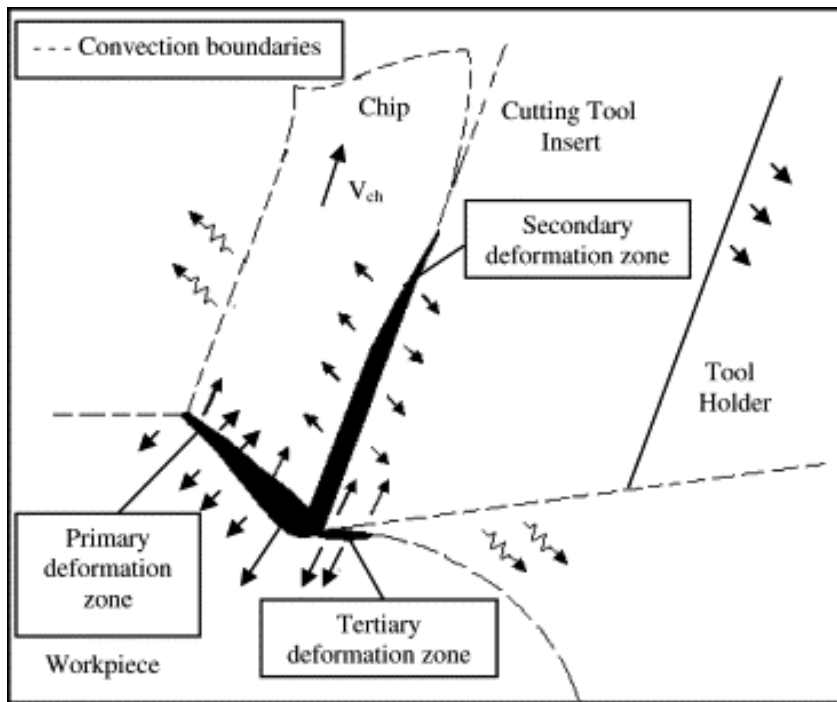


The Four Domains of Finishing Processes



We need to start with the thermal domain, since the behavior of all other domains is strongly dependent on temperature!

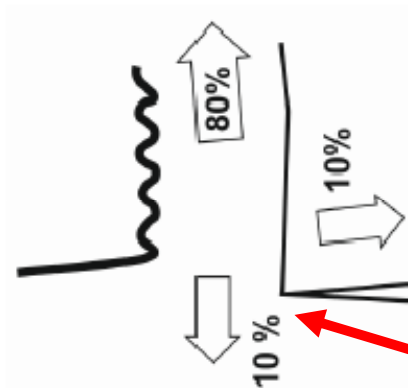
Heat in Metal Cutting



80-90% of consumed power is converted to HEAT (random particle movement, leading to increased temperatures)

In roughing, ~10% of total heat goes into the workpiece

In finishing, ~80-90% of the total heat goes into the workpiece!
(the remainder into the tool)



This is where to sub-surface is heated!

The Origin of Heat in Finishing

Plastic work involves internal friction

External (normal) **friction** leads to significant (rubbing) heat on the tool flank

- Combine these two zones into an equivalent heat source to simplify calculations!

Recall: $\mu = F_{\text{friction}} / F_{\text{normal}}$

In the rubbing (sliding/ploughing) regime, i.e. finishing: $F_{\text{friction}} \approx F_{\text{cutting}}$

- Assuming heat is due to (overall/average) friction, we can also assume that the **tangential/cutting force is the cause of heat generation!**
- Note that this assumption fails as soon as a meaningful amount of material removal occurs (i.e., this will not work for roughing without further adjustments!)

From Forces to Temperatures

Units of forces: Newtons (N)

Units of temperature: Kelvin/Celcius (K or °C)

-> a macroscopic/average expression of heat in a specific material!

Units of heat (energy and work): Joules (J), where $J = N \cdot m$

What do we need in addition to friction force to get units of Joules?

- **(Frictional) Contact length** in meters!
- *Note that this length is not just the tool/chip contact length, but rather the entire frictional interface, including the plastic zone!*

We will also need some material specific properties, e.g. *specific heat, thermal conductivity, density*, etc. as well as the effect of *cutting speed*.

- These effects can be included through the non-dimensional **Péclet number**!

How to get to 2D (plane strain) temperature profiles

1) Determine maximum/flash temperature on surface (*from dimensional analysis and Peclet number*)

2) Determine normalized surface temperature distribution

2.2) Scale normalized distribution using the Flash Temperature from step 1)!

3) Approximate 1D sub-surface temperature distribution for each point below surface temperature to obtain full 2D profile

3.2) Using the 1D (depth) profile below the max/flash temperature, we can also estimate the maximum tensile residual stresses!

Analytical Modeling of Thermal Effects: Péclet Number

Strongly affected by **material properties**, which dynamically **change** with pressure, temperature, etc.

In sliding contact, the Péclet number (Pe) is a key measure defining the shape of the thermal field:

$$Pe = \frac{v_c b \rho c_p}{2 k}$$

e.g, for $v_c = 120 \text{ m/min}$

Pe(Ti6Al4V) = 6.80 *For example:*

Pe(IN 718) = 6.26

Pe(52100) = 2.13

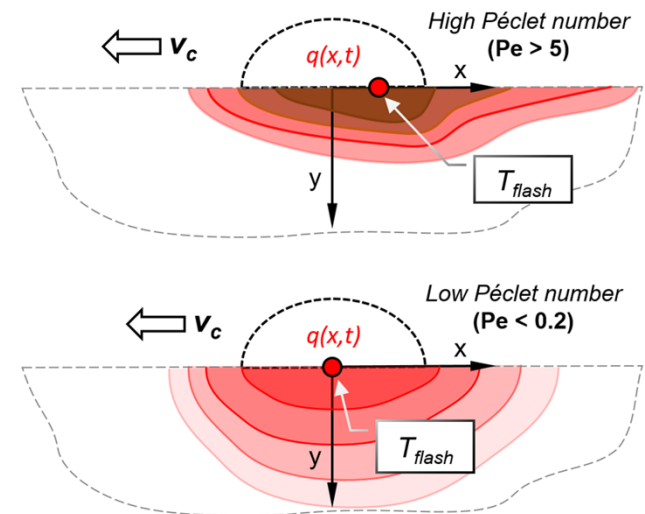
Pe(WC) = 0.57

$$\begin{aligned} v_c &= \text{cutting speed } \left(\frac{\text{m}}{\text{s}} \right) \\ b &= \text{half width of contact } (\text{m}) \\ \rho &= \text{density } \left(\frac{\text{kg}}{\text{m}^3} \right) \\ c_p(f_{xn}(T,P)) &= \text{specific heat } \left(\frac{\text{J}}{\text{kg K}} \right) \\ k(f_{xn}(T,P)) &= \text{thermal conductivity } \left(\frac{\text{W}}{\text{m K}} \right) \end{aligned}$$

$$\rho_{\text{Ti6Al4V}} = 4500 \left(\frac{\text{kg}}{\text{m}^3} \right)$$

$$c_p(\text{Ti6Al4V}) = 1.71 * 10^{-6} T^3 - 0.00173 T^2 + 0.60 T + 536 \left(\frac{\text{J}}{\text{kg C}} \right)$$

$$k_{\text{Ti6Al4V}} = 1.12 * 10^{-5} T^2 + 0.00982 T + 6.4 \left(\frac{\text{W}}{\text{m C}} \right)$$



Analytical Modeling of Thermal Effects: Flash Temperature

The 'flash temperature' is the maximum temperature the tool/workpiece experiences during frictional sliding

Heat source intensity (line source in plane strain) : $q = \frac{F_c v_c}{2b}$ in units of W/m

Transient time of heat source (characteristic time): $t = \frac{2b}{v_c}$

The definition and calculation of T_{flash} changes, depending on the Peclet regime:

For **Pe < 5** (slow/equilibrium conditions): $T_{flash} = C_4 * 0.159 * \frac{F_c}{\rho c_p w b}$

- Where $C_4 = f(Pe) = 0.00527 Pe^3 - .192 Pe^2 + 2.39 Pe$ (see [this link](#) for more information from Stachowiak's book)

For **Pe > 5** (fast/transient conditions): $T_{flash} = 0.399 * \frac{F_c v_c}{k w} \sqrt{\frac{k}{\rho c_p v_c b}}$

- In these equations, **w** represents the width (into the plane); in orthogonal machining this is the width of the tool/workpiece, where typically w is about 1-10mm and **F_c** is between 10-1000 N

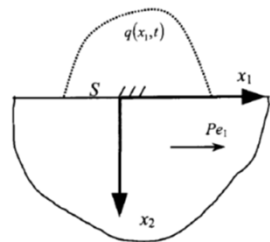
Analytical Modeling of Thermal Effects: Surface Temperatures

The yet-to-be-scaled temperature distribution on the surface is given by a quasi-static solution to the transient heat equation (relying on the modified Bessel Equations K_0 and K_1):

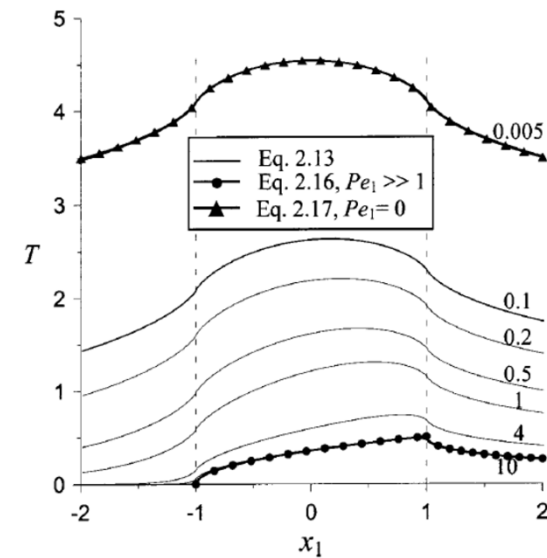
$$T_{ss} = \begin{cases} \left\{ \begin{aligned} &\left(\frac{x}{b} + 1 \right) e^{P_+} [K_0(-P_+) - K_1(-P_+)] \\ &+ \left(1 - \frac{x}{b} \right) e^{P_-} [K_0(-P_-) - K_1(-P_-)] \end{aligned} \right\} & \frac{x}{b} \leq -1 \\ \left\{ \begin{aligned} &\left(\frac{x}{b} + 1 \right) e^{P_+} [K_0(P_+) + K_1(P_+)] \\ &+ \left(1 - \frac{x}{b} \right) e^{P_-} [K_0(-P_-) - K_1(-P_-)] \end{aligned} \right\} & -1 \leq \frac{x}{b} \leq 1 \\ \left\{ \begin{aligned} &\left(\frac{x}{b} + 1 \right) e^{P_+} [K_0(P_+) + K_1(P_+)] \\ &+ \left(1 - \frac{x}{b} \right) e^{P_-} [K_0(P_-) + K_1(P_-)] \end{aligned} \right\} & 1 \leq \frac{x}{b} \end{cases}$$

where $P_{\pm} = \frac{Pe(\frac{x}{b} \pm 1)}{2}$

Adapted/modified from: Liu et al., **Solutions for Temperature Rise in Stationary/Moving Bodies Caused by Surface Heating With Surface Convection**, Journal of Heat Transfer, Transactions of the ASME, Vol. 126, 2004, pp 776-785



(b) Half-plane



Analytical Modeling of Thermal Effects: Sub-Surface Temperatures

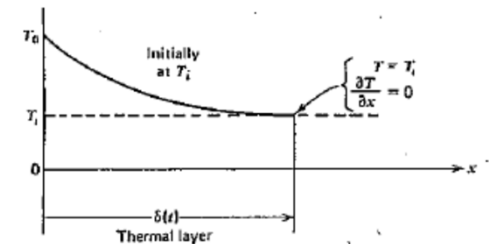
Once the relative surface temperature profile and flash temperature are known, the sub-surface temperature distribution can also be calculated using the appropriate relationship (read Chapter 9, Ozisik's Heat Conduction book)

Along a single axis (i.e. 1D), the relevant solution to the heat equation is the well-known one for a conducting, semi-infinite half-plane:

$$\frac{\partial^2 T(z, t)}{(\partial x^2)} = \frac{1}{\alpha} \frac{\partial T(z, t)}{\partial t}, \quad \text{in } z > 0, t > 0$$

$$T(z, t) = T_0, \quad \text{at } z = 0, t > 0$$

$$T(z, t) = T_i, \quad \text{for } t = 0, \text{ in } z \geq 0$$



Using the integral method, we obtain:

$$T(z, t) = \left(1 - \operatorname{erf} \frac{z}{\sqrt{4 \alpha t}} \right) T_{flash} + T_{bulk},$$

where $\alpha = \frac{k}{\rho c_p}$, and $t = \frac{2b}{v_c}$

Notes on Calculating 1D Depth Profiles

We will want to set up our calculator by starting with depth normalized by the half width of contact, i.e. z/b for generally applicable scaling (i.e., we can adapt to any new value of b rather than create a new calculator for each specific b)

Our z/b depth values will vary from 0 (top surface) to no more than 4

In order to calculate the sub-surface profile we need to convert our normalized z/b range back to z by multiplying b ; this may seem redundant, but normalization is necessary for generally applicable scaling and multiplication by b is necessary, since the error function contains other explicit values, whose units will only cancel out properly whenever z has units of depth (e.g., meters or microns).

What to do with the 2D temperature profile data?

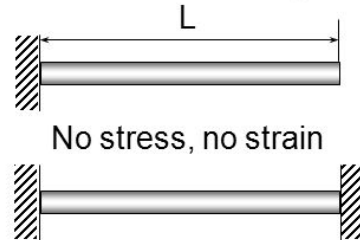
Using a map of temperature and the yield strength of a material as a function of temperature, as well as thermal expansion coefficient data, we can calculate **tensile residual stresses**!

When the **equivalent stress due to thermal expansion** exceeds the yield strength at that temperature, permanent (plastic) deformation occurs!

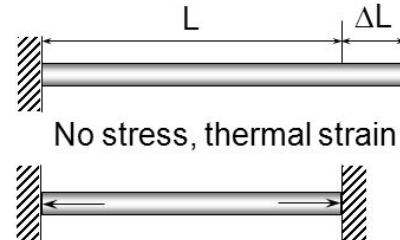
The retained (plastic) strain is stored as a ***tensile residual stress due to thermal effects***

Thermal (expansion) Stresses

- Temperature change causes thermal strain



(a) at $T = T_{\text{ref}}$



Thermal stress, no strain

(b) at $T = T_{\text{ref}} + \Delta T$

- Constraints cause thermal stresses
- Thermo-elastic stress-strain relationship

$$\sigma = E(\varepsilon - \alpha\Delta T)$$

Thermal expansion coefficient

$$\varepsilon = \frac{\sigma}{E} + \alpha\Delta T$$

Development of tensile residual stresses

Tensile residual stresses (RS) develops if and when equivalent stress due to thermal expansion exceeds the yield strength at a given temperature:

iff $\sigma_{eq} > YS$

$$\sigma_{RS} = \sigma_{equivalent}(T) - YS(T)$$

Titanium alloy 6Al-4V example:

$$YS(T)_{Ti6Al4V} \approx 0.005752 * T^4 - 10.67 T^3 + 5597 T^2 - 1.79 * 10^6 T + 1.056 * 10^9 \text{ {Pa}}$$

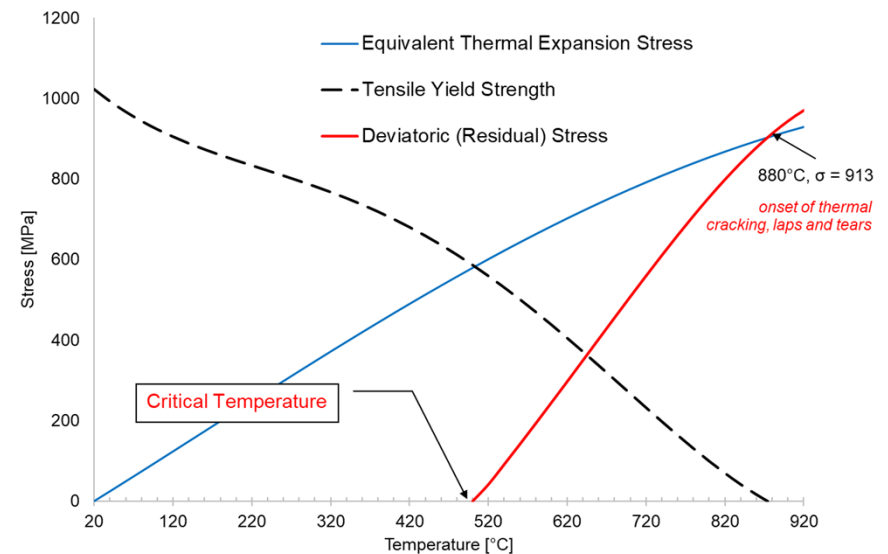
$$E(T)_{Ti6Al4V} \approx -4 * 10^7 T + 1 * 10^{11} \text{ {Pa}}$$

$$\alpha_T (Ti6Al4V) \approx (1.23 * 10^{-6} * T^2 + 2.87 * 10^{-3} T + 4.57) * 1.8 * 10^{-6} \text{ {m/m K}}$$

$$\nu_{Ti6Al4V} \approx 0.32$$

Using the above information, you should be able to generate an approximately parabolic function of residual stress as a function of temperature for Ti-6Al4V!

$$\sigma_{equivalent}(T) = \frac{E_w(T) \alpha_{T,w} (T_f - T_0)}{1 - \nu} \geq YS(T)$$



What to do before the next discussion session:

Prepare a spreadsheet or Matlab, Python, etc. program that calculates and graphs:

- 1) A 2D map of temperature based on relevant material property inputs, as well as contact length, cutting/friction force and cutting speed as key inputs
- 2) An estimation of residual stress as a function of depth (may be normalized by the contact length b , or in units of meters)
 - *Hint: you will technically only need the flash/max temperature for this part, and the graph will represent a transition from 'temperature space' to 'residual stress space'*

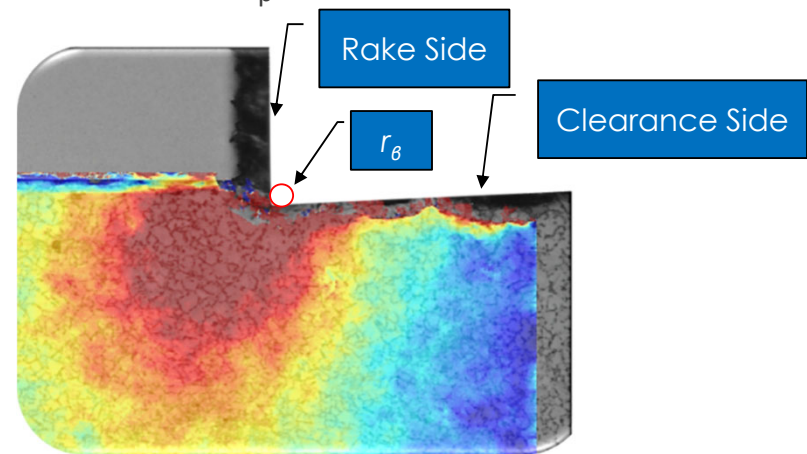
You can build your program with the data provided for Ti-6Al4V, but try to keep the architecture open to be able to add additional materials in the future (i.e., allow for material properties to be changed easily)

Connecting the **Thermal Model** to Machining Parameters

Key variables: b , v_c , F_c , workpiece physical properties (k , ρ , k , c_p)

Process parameters:

- Cutting speed (v_c)
- Uncut chip thickness/feed (h)
- Cutting edge radius/hone (r_θ)
- Rake angle (γ)
- Clearance angle (α)



Key question: How do we find b and F_c for each combination of process parameters???

Connecting the Thermal Model to Machining Parameters

Recall:: We built the model base on assumption that heat in metal cutting is the result of friction, both internal (severe plastic deformation) and external (sliding)

We have two zones:

1. Internal friction zone (primary shear deformation)
2. External friction zone (clearance/springback)

For zone 2, the length of external contact is given as: $l_{ex} = \frac{h_{min}}{\tan \alpha}$, and for $h_{min} \sim \frac{r_\beta}{2}$,
$$l_{ext} = \frac{r_\beta}{2 \tan \alpha}$$

For zone 1, we need to carefully consider how heat is generated in the shear plane

Connecting the Thermal Model to Machining Parameters

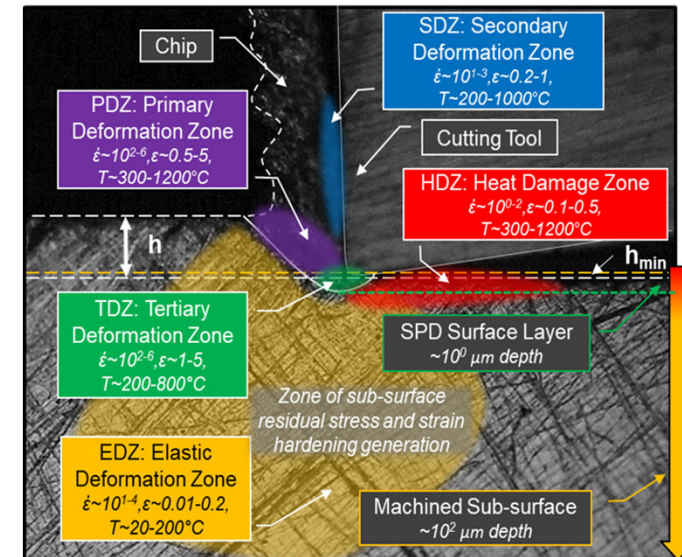
For zone 1, we need to carefully consider how heat is generated in the shear plane

Step 1: The shear plane has a shear angle Φ , which is approximately equal to 35 degrees for most metals for a rake angle γ of zero, and changes approximately linearly with rake angle ($\gamma = +5 \rightarrow \Phi = 40$, $\gamma = -5 \rightarrow \Phi = 30$), so: $\Phi = 35 + \gamma$
The length of the shear plane is thus given as $l_s = \frac{h}{\sin \Phi}$

Step 2: Since the plastic heat generation in the shear plane encompasses 80% of the total heat generated, we need the thermal (Peclet) number of the shear plane. Instead of $2b$, we use l_s and get: $Pe_{shear} = \frac{v_c l_s \rho c_p}{4 k}$

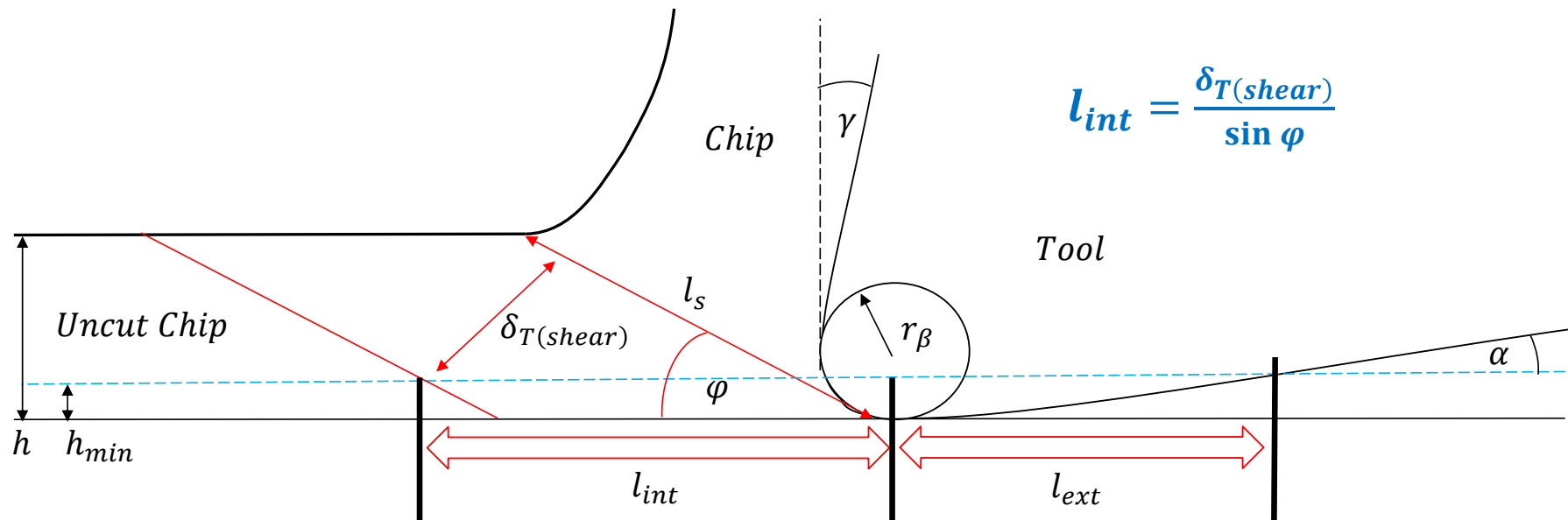
Step 3: We can calculate the thermal layer thickness of the shear plane, i.e., the depth to which the heat generated in the shear plane extends away from the shear plane.

$$\delta_{T(shear)} = \sqrt{v_c / Pe_{shear}}$$



Connecting the **Thermal Model** to Machining Parameters

We can now estimate the dimensions of zone 1, the shear plane heat source



Connecting the Thermal Model to Machining Parameters

Recall:: We built the model base on assumption that heat in metal cutting is the result of friction, both internal (severe plastic deformation) and external (sliding)

We have two zones:

1. Internal friction zone (primary shear deformation) $l_{ext} = \frac{r_\beta}{2 \tan \alpha}$
2. External friction zone (clearance/springback) $l_{int} = \frac{\delta T(\text{shear})}{\sin \phi}$

Now we can determine the effective half width of thermal contact, b :

$$2b = l_{ext} + l_{int}, \quad \text{so: } b = \frac{l_{ext} + l_{int}}{2} \rightarrow \text{fxn}(r_\beta, h, \alpha, \gamma)$$

Connecting the Thermal Model to Machining Parameters

To determine cutting forces, we can simply measure the forces for a real cutting process.

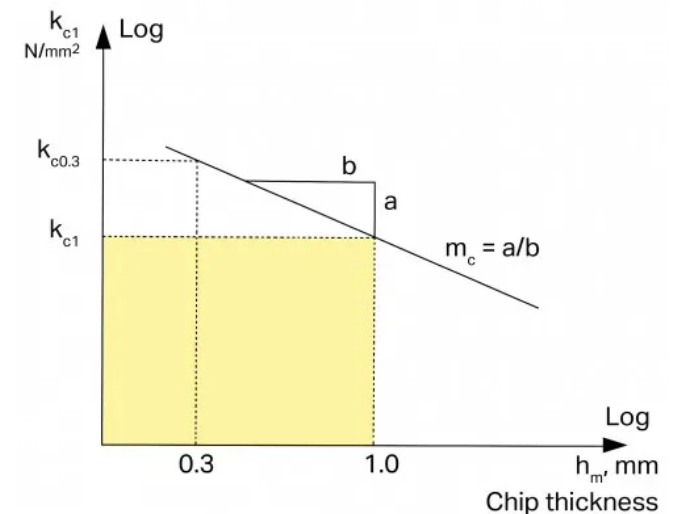
- The spreadsheet 'Cutting Force Data' on Canvas contain typical measurements, that can be readily fit with a polynomial regression for the cutting force, or a power law regression for the specific cutting force.
- The specific cutting force is often more convenient
- $K_c = F_c / (h * w)$, where w is the width of cut
 - In orthogonal experiments, w is typically 3mm
- The specific force follows a power law relationship

$$k_c = k_{c1} * h^{-m_c} * \left\{ \frac{100 - \gamma}{100} \right\}$$

k_{c1} is the specific force at h and w of 1mm

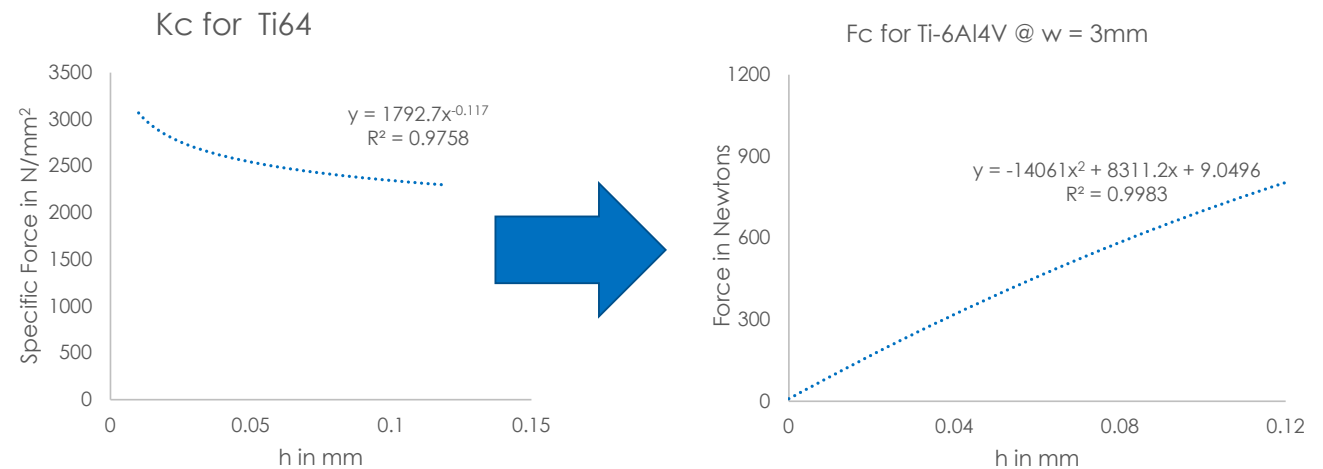
m_c is the strain hardening exponent

To convert $k_c(h)$ back to F_c , we simply need to multiply k_c by h and w



Connecting the **Thermal Model** to Machining Parameters

Once we have a cutting force relationship, we can determine the forces as a function of feed (and/or rake angle):



Connecting the Thermal Model to Machining Parameters

The final piece of information we need is the heat partition of the percentage of total heat (total F_c) that enters the workpiece, R

Although R changes with process, we can estimate it as:

$$R = 0.6 * Pe_{shear}^{-0.4}$$

Homework: Based on all of this information, calculate the flash temperature for Ti-6Al4V as a function of cutting speed v_c (0.05 to 5 m/s) and for $h = 0.02, 0.04$, and 0.08 mm

Credit: DFG, Brinksmeier et al.,

